

Response to Sensei's Comments

How the Paper Was Updated

This note lists each comment and what was done about it.

1 Question 1: More Fine-Tuning Scenarios

The comment: The findings came from only one fine-tuning scenario. Another tuning campaign might behave differently, and showing that would be an important conclusion.

What I did: Instead of one retraining, I ran **15 retrainings** of C1 (the detector). I changed three things:

- **How much data:** the full Singapore training set (119 images), half of it (60), and a quarter (30).
- **How long we train:** 5, 10, and 20 epochs.
- **Random seed:** every setting repeated 3 times, so we can tell a real effect from luck.

C2 (prediction) and C3 (planning) were never touched. Only C1 changed. Each run was measured in the same way as before.

The result: The entanglement is a range, not one number.

- The retraining *always* moved the driving plan, but by very different amounts: from **1.33 m to 7.80 m**.
- The planner got *worse* in 11 runs and *better* in 4. So “retraining hurts the planner” is not a universal rule.
- More fine-tuning data caused a bigger downstream change. The full-data runs moved the plan by 6.12 m on average, the smaller sets by about 3 m.

This variety is itself the conclusion: different tuning campaigns really do produce different entanglements. It is in the paper as **Finding F6**, with a full 15-row table.

2 Question 2: Evaluating CARA

The comment: CARA was only presented as an idea. Can its effectiveness be evaluated, at least to some extent?

What CARA is, in one line: Before we accept a retrained detector, CARA compares it against the current one and decides: *admit* it, or *hold* it for inspection.

How I evaluated it: CARA has a cheap front-end called the *interface-drift screen*. For each retrained detector, it looks only at that detector’s own outputs and measures how different they are from the old detector’s outputs. It uses four simple things: how many objects are found per image, how confident the detector is, where the boxes sit, and which classes appear. These are combined into one *drift score* between 0 and 1.

Table 1 shows the full measurement on all 15 runs. It is sorted by drift score, and the last column marks whether the run really did damage the planner.

I asked two questions of this data.

Table 1: The complete evaluation data. *Drift* is computed from the detector’s outputs only. *Plan move* is how far the driving plan actually shifted. *Harmful* means the planner really got worse.

Run	Drift	Plan move (m)	Δ_3	Harmful?
full, 10 ep, s1	0.582	7.80	+0.68	harmful
full, 10 ep, s2	0.528	7.40	+1.34	harmful
full, 10 ep, s3	0.527	3.17	+0.26	harmful
full, 20 ep, s2	0.525	3.17	+0.26	harmful
full, 20 ep, s3	0.517	3.17	+0.26	harmful
quarter, 10 ep, s2	0.478	3.29	−0.20	harmless
full, 20 ep, s1	0.456	2.59	+0.24	harmful
half, 10 ep, s3	0.420	2.59	+0.24	harmful
half, 10 ep, s2	0.409	3.17	+0.26	harmful
full, 5 ep, s2	0.404	1.33	−0.00	harmless
full, 5 ep, s3	0.399	1.33	−0.00	harmless
quarter, 10 ep, s1	0.380	3.17	+0.26	harmful
full, 5 ep, s1	0.352	2.59	+0.24	harmful
quarter, 10 ep, s3	0.308	3.93	−1.04	harmless
half, 10 ep, s1	0.300	3.17	+0.26	harmful

Evaluation 1: does a high drift score predict a big downstream problem? Comparing the Drift column against the Plan-move column gives a rank correlation of **0.41**. There is a real signal, but it is loose. It works best at the top of the table: the two runs that moved the plan the most (7.80 m and 7.40 m) are also the two highest-drift runs. In the middle of the table the two columns disagree more often.

Evaluation 2: does it work as an admit/hold decision? Of the 15 runs, 11 were harmful and 4 were harmless. Reading down the Harmful column shows what a threshold can and cannot do: most harmful runs sit high, but one harmless run (quarter, 10 ep, s2) sits at 0.478, and one harmful run (half, 10 ep, s1) sits at the very bottom at 0.300. Overall the score ranks a harmful run above a harmless one about **70% of the time** (AUROC 0.70). Table 2 shows the two thresholds.

Table 2: Admit/hold decisions at two thresholds.

Threshold	Harmful caught	Harmless admitted	Wrong holds
Strict (0.25)	11 of 11 (100%)	0 of 4 (0%)	4
Balanced (0.41)	8 of 11 (73%)	3 of 4 (75%)	1

The strict threshold holds every run. Nothing bad gets through, but nothing good is ever admitted either, so it is safe and useless at the same time. The balanced threshold is the usable one: it catches about three-quarters of the harmful runs while correctly admitting three-quarters of the harmless ones, with only one wrong hold.

Which signal does the work: The drift score combines four signals, and they do not contribute equally. Averaged over the 15 runs, the horizontal box position matters most (0.81), then the confidence (0.58), then the vertical box position (0.38), then the number of objects per image (0.36). The class mix barely matters (0.07). In other words, retraining mainly changes *where* and *how many* objects are detected, not *which* classes.

Limits of this evaluation: This is a proof of concept, not a proven rule. The balanced threshold was chosen using the same 15 runs it is judged on, so there is no separate test set, and there are only 3 test scenes. The collision-based part of CARA could not be tested, because none of the runs caused a collision. All of this is stated in the paper.

Both evaluations appear in the paper, with a figure for Evaluation 1 and a table for Evaluation 2.

3 The Main New Result: Real Entangled Enhancement

The problem we had: Entangled enhancement means: *the upstream part gets better on its own score, and yet the whole system gets worse*. In all 15 runs above, C1’s own score always went *down*. So we could show related effects, but never the real thing.

Why the score kept dropping: It was not the pipeline. It was the data. The 119-image Singapore set is dominated by the `car` class and the other classes are rare, so a small fine-tune overfits to cars and the overall mAP falls.

What I did: I rebuilt the training set so the classes are more balanced: images containing rare classes are repeated, so the class mix flattens. The validation set was left completely untouched, so the scores stay comparable to every earlier run. Then I ran **12 more retrainings** (2 balance strengths $\times$ 2 training lengths $\times$ 3 seeds).

The result: It worked.

- C1’s own score went **up** in **6 of the 12** runs. The balancing removed the accuracy penalty, as intended.
- In **3 of those runs, the planner got worse at the same time**. That is exactly *entangled enhancement*: the detector improved on its own metric, and the system still degraded.

Table 3: The three runs with real entangled enhancement. δ_1 = detector’s own score change (positive = better); Δ_3 = planner error change (positive = worse).

Run	δ_1	Δ_3	Entangled enhancement?
balance 0.5, 20 epochs, seed 1	+0.031	+0.24	yes
balance 1.0, 20 epochs, seed 3	+0.021	+0.24	yes
balance 1.0, 20 epochs, seed 2	+0.012	+0.24	yes

All three happen at the longer training length (20 epochs), under both balance settings and across two different seeds, so it is not one lucky run.

Why this matters: Each of these three retrained detectors would have *passed* a normal component-level check, because the detector’s own score improved. Yet each one made the driving output worse. This is the strongest argument for why a method like CARA is needed, and it is now the paper’s central empirical claim (**Finding F7**).

4 Structure and Writing Changes

Paper structure: The paper now follows the suggested technical-paper order exactly:

1. Introduction
2. System Model (*including the definition of entangled enhancement*)
3. CARA (the proposed technique)
4. Experiment Setup
5. Evaluation Results
6. Discussion
7. Related Work
8. Conclusion

CARA was moved out of the back of the paper. It now has its own section *before* the experiments, so the proposed technique is presented first and then evaluated. Related Work was moved to the back.

Introduction shortened: The end-to-end (E2E) discussion had grown too long. It is now two sentences that simply say why we use a modular pipeline. The detailed E2E material stayed in Related Work, where it belongs.

Writing: Two things were fixed throughout the paper:

- **Long sentences split.** Sentences that joined several different topics with “:” or “;” were divided into shorter ones. For example, the sentence about C1, C2 and C3 is now four separate sentences.
- **Terse sentences explained.** Sentences that were hard to follow now have an introduction first. For example, “This paper supplies that measurement” now says *what* measurement before saying it, and “The three isolated-mode rows are exactly zero-delta” now first explains that this is a validity check, not a result.